Section 2.3-4

Solving systems of linear equations by elementary row operations / row echelon forms

Ex 1:

(1) $x_1 + x_2 + x_3 = 3$

(2) $x_2 + 2x_3 = -1$

(3) $x_3 = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

Row(2) - 2 Row(3) $\Rightarrow x_2 = -3$ (4)

Plug (3) & (4) into (1) $\Rightarrow x_1 = 3 - x_2 - x_3 = 3 + 3 - 1 = 5$

Def A matrix is in row echelon form if

a) all zeros rows are near the bottom

b) the first nonzero entry of a row, called the leading entry, is always to the right of the leading entry of the row above it.

Examples:

$$\begin{bmatrix} x & x & x & x & x \\ 0 & x & x & x & x \\ 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & 0 & x \end{bmatrix} \qquad \begin{bmatrix} x & x & x & x & x & x \\ 0 & 0 & 0 & x & x & x \\ 0 & 0 & 0 & 0 & 0 & x \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Elementary Operation

- Exchange of two rows
- Multiplication of a row by a constant
- Adding of two equations / rows. (Here by addition we mean c_1 row i + c_2 row j)

Example 1:

$$\begin{aligned} x_1 - x_2 + x_3 - x_4 &= 2 \\ x_1 - x_2 + x_3 + x_4 &= 0 \\ 4x_1 - 4x_2 + 4x_3 &= 4 \\ -2x_1 + 2x_2 - 2x_3 + x_4 &= -3 \end{aligned}$$

$\Rightarrow$ Augmented matrix $[A|b]$

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 1 & -1 & 1 & 1 & 0 \\ 4 & -4 & 4 & 0 & 4 \\ -2 & 2 & -2 & 1 & -3 \end{array}\right] \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - 4R_1 \\ R_4 = R_4 + 2R_1 \end{array}$$

$$\Rightarrow \underbrace{\left[\begin{array}{cccc} 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & -1 \end{array}\right]}_{A} \underbrace{\left|\begin{array}{c} 2 \\ -2 \\ -4 \\ 1 \end{array}\right]}_{b} \xrightarrow[R_4 = R_4 + R_2/2]{R_3 = R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$x_4 = -1 \Rightarrow$ Back to 1st equation

$$x_1 - x_2 + x_3 = 2 + x_4 = 1$$

Take $x_3 = \lambda_2 \quad x_2 = \lambda_1$

$$x_1 = 1 + \lambda_1 - \lambda_2$$

Solution $\Rightarrow$

$$\begin{aligned} x_1 &= 1 + \lambda_1 - \lambda_2 \\ x_2 &= \lambda_1 \\ x_3 &= \lambda_2 \\ x_4 &= 1 \end{aligned} \qquad \lambda_1, \lambda_2 \in \mathbb{R}.$$

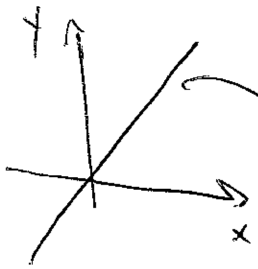
We can re-write as

particular solution for $Ax=b$

general solution for $Ax=0$

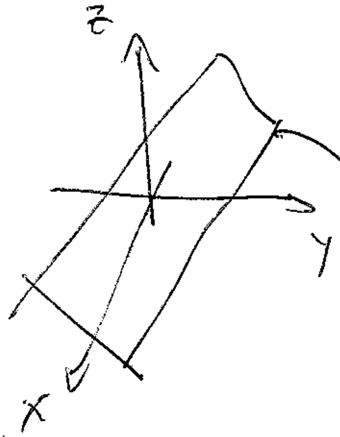
$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + \lambda_1 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\text{general solution for } Ax=b}$$

Vector Space and Vector Subspace



xy-plane is a vector space

line is a subspace



$xyz \Rightarrow \mathbb{R}^3$ is a vector space

plane is a subspace.

Groups: Consider a set G, and an operation $G \times G \to G$, then $(G, \otimes)$ is called a group if

1) $\forall x, y \in G \quad x \otimes y \in G$
2) associativity $\quad x \otimes (y \otimes z) = (x \otimes y) \otimes z$
3) neutral element $\quad \exists e \in G$ such that $\forall x, \quad x \otimes e = x$, $\quad e \otimes x = x$
4) Inverse element $\quad \forall x \in G, \underbrace{\exists y \in G}_{\text{denoted by } x^{-1}}$ s.t. $x \otimes y = e$, $\quad y \otimes x = e$

If in addition $x \otimes y = y \otimes x$, we call it an Abelian Group.

Ex 1: a) $(\mathbb{R}, +)$ is an Abelian Group
reals

b) $(\mathbb{Z}, +)$ is an Abelian Group
integers

c) $(\mathbb{R}, \cdot)$ is not a group $\rightarrow$ No inverse for 0.

$(\mathbb{R} \setminus \{0\}, \cdot)$ is an Abelian group

d) $(\mathbb{R}^n, +)$ are Abelian for any n, with "+" defined as $(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$

e) $(\mathbb{R}^{m \times n}, +)$ is an Abelian group
$m \times n$ matrices

f) (set of all invertible $n \times n$ matrices, matrix product) is a group (not Abelian)

Question to think:

$(\mathbb{R}^3$, cross product$) \Rightarrow$ group?

$$\left(\underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}_{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}} \right) \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \overset{?}{=} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \left(\underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}_{\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}} \right)$$

you have

$$LHS = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \qquad RHS = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Answer: No → Not associative

Vector Space

A real valued vector space $V = (\underbrace{V}_{\text{set}}, \underbrace{+, \cdot}_{\text{two operations}})$

with $+ : V \times V \to V$

$\cdot : \mathbb{R} \times V \to V$

and

1. $(V, +)$ is an Abelian Group
2. $\forall \lambda \in \mathbb{R}$, $x, y \in V$

   $\lambda \cdot (x + y) = \lambda \cdot x + \lambda \cdot y$

   $\forall \lambda, \alpha \in \mathbb{R}$, $x \in V$

   $(\lambda + \alpha) \cdot x = \lambda \cdot x + \alpha \cdot x$

   distributivity
3. $\lambda \cdot (\alpha \cdot x) = (\lambda \alpha) \cdot x$ → Associativity
4. $\forall x \in V$, $1 \cdot x = x$ → Neutral Element "1"

Example:

1) ($\mathbb{R}^n$, vector addition, scalar-vector product) is a vector space

2. $(\mathbb{R}^{m\times n}$, matrix addition, scalar-matrix product$)$ is also a vector space

3. (polynomials, polynomial addition, scalar-pol. product) is a vector space

4. (set of cont. functions, standard function, scalar mult.) is a vector space.

Vector Subspace

$V = (\mathcal{U}, +, \cdot)$ is called a subspace of $\mathcal{V}$, if

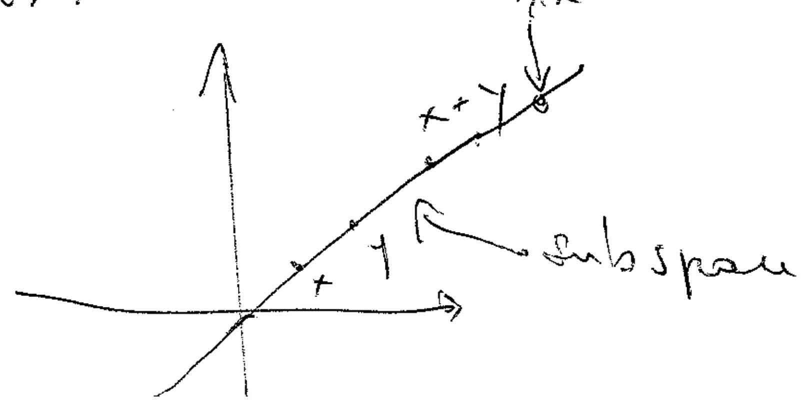
a) $\mathcal{U}$ is a subset of $\mathcal{V}$

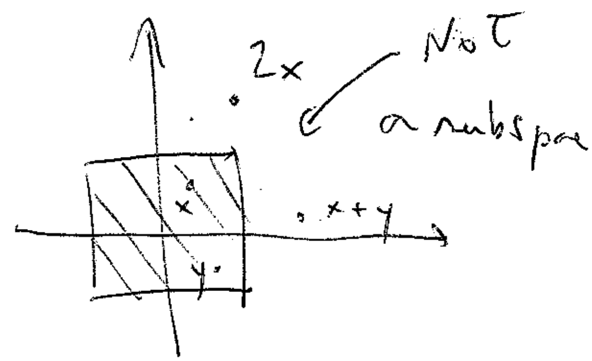
b) V is a vector space itself.

$\Leftrightarrow$ (1) for any $x \in \mathcal{U}$, $\lambda \in \mathbb{R}$ $\quad \lambda x \in \mathcal{U}$

(2) for all $x, y \in \mathcal{U}$, $\quad x + y \in \mathcal{U}$.

Ex:





Remark:

a) Take $A \in \mathbb{R}^{m \times n}$

The set $\{x \in \mathbb{R}^n ; Ax = 0\}$ with $+$ and $\cdot$ is a subspace of $\mathbb{R}^n$

b) The set of all differentiable functions is a subspace of the set of all cont. functions.